

Database Metadata

Course: DLBDSPBDM01 – Build a Data Mart in SQL; **Programme:** B.Sc. Computer Science; **Author:** Zubair Ahmed Khan; **Matriculation Number:** 92127983; **Tutor:** Prof. Dr. Anna Androvitsanea; **Submission Date:** 2026-07-29

1. Identification

Attribute	Value
Database Name	AirbnbDB
Purpose	Coursework data mart modelling an Airbnb-style short-term rental platform, supporting users, guests, hosts, accommodations, bookings, reservations, payments, reviews, complaints, notifications, amenities, locations, and listings.
Version	1.0

2. Technical Environment

Attribute	Value
Database Management System	MySQL Server 8.0
Client Tool	MySQL Workbench 8.0
SQL Language	SQL (DDL and DML)
Storage Engine	InnoDB (default)
Character Set	utf8mb4 (default)
Server Platform	Cross-platform (Windows, macOS, Linux)

3. Schema Statistics

Attribute	Value
Number of Tables	21
Number of Entities	21 (one-to-one with tables)
Number of Relationships	One recursive (Employee–Manager); one ternary (Booking–Payment–Reservation); one many-to-many (Place–Amenity through AmenityListing); plus a network of one-to-many and one-to-one links
Primary Keys	21 — one per table, integer, named <code><TableName>ID</code>
Foreign Keys	Implemented across dependent tables; reinforced by <code>ALTER TABLE</code> for late-bound references
Minimum Records per Table	20
Total SQL Statements	70+ (CREATE, ALTER, INSERT, SELECT, SHOW)
Estimated Storage Footprint	Less than 1 MB (small data volume; exact size depends on server configuration and BLOB allocation)

4. Normalisation and Integrity

Attribute	Value
Normalisation	1NF, 2NF, 3NF (First, Second, and Third Normal Form)
Primary Key Strategy	Surrogate integer key per table
Foreign Key Strategy	Declarative <code>FOREIGN KEY</code> constraints; orphan rows rejected
Atomicity	All attributes hold single-valued data; no repeating groups
Dependency Hygiene	Non-key attributes depend on the whole primary key, not on other non-key attributes
Referential Integrity	Enforced through foreign-key constraints
Domain Integrity	Enforced through data types (<code>DECIMAL(10,2)</code> , <code>DATE</code> , <code>DATETIME</code> , <code>VARCHAR</code> , <code>BLOB</code>)

5. Testing

Attribute	Value
Number of Test Cases	3
Test Case 1	Retrieval of complete booking information (multi-table join over Booking, Guest, User, Accommodation)
Test Case 2	Calculation of host accommodation statistics and income (aggregation over Accommodation, Booking, Host, User)
Test Case 3	Retrieval of complete accommodation information (join over Accommodation, Host, User)
Verification Method	Multi-table JOIN statements returning non-empty result sets; <code>SELECT COUNT(*)</code> per table returning ≥ 20

6. Database Features

Feature	Description
Referential Integrity	Foreign-key constraints reject inserts and updates that would break parent–child links
Dummy Data	At least twenty realistic rows per table, sufficient for non-trivial joins and aggregations
Recursive Relationship	Employee references itself through ManagerID to model an organisational hierarchy
Ternary Relationship	Booking, Payment, and Reservation form a triple association representing the full reservation lifecycle
Many-to-Many Link	AmenityListing resolves the many-to-many between Place and Amenity
Schema Evolution	<code>ALTER TABLE</code> statements add new columns and foreign keys without data loss
Consistency	All non-key attributes depend on the primary key only (3NF); update anomalies are prevented
Scalability	The schema accepts new users, hosts, accommodations, and bookings without structural change
Portability	Pure SQL, no proprietary extensions; runs on any MySQL 8.0 server

7. Repository and Distribution

Attribute	Value
GitHub Repository	https://github.com/zbr-khn/SQL
Submission Folder	DLBDSPBDM01_Final_Submission/
SQL File Location	DLBDSPBDM01_Final_Submission/SQL/airbnb.sql
Documentation	Abstract, Installation Manual, SQL Documentation, Database Metadata, Submission Checklist
Total Deliverables	Combined PDF portfolio + ZIP folder + individual PDFs

8. Inventory of Tables

#	Table	Purpose
1	User	Platform participants (guests and hosts)
2	Guest	Specialisation of User for booking stays
3	Host	Specialisation of User for listing accommodations
4	Accommodation	Property listed by a host
5	Booking	Reservation of an accommodation by a guest
6	Payment	Financial settlement of a booking
7	Reservation	Confirmed reservation linking booking and payment
8	Review	Guest-to-host or host-to-guest review
9	Complaint	Issue raised by a guest or host
10	Notification	System message directed to a user
11	Location	Geographic address (country, region, city, street, post code)
12	HouseRules	Set of rules associated with a place
13	Place	Physical description of a property (rooms, size, max guests)
14	Amenity	Catalogue of amenities (WiFi, AC, pool, etc.)
15	AmenityListing	Many-to-many link between Place and Amenity
16	Media	Photos and other media assets
17	Listing	Public listing of a place with cover photo and description
18	Employee	Back-office staff with a self-referencing manager relationship
19	Language	Catalogue of spoken languages

#	Table	Purpose
20	UserLanguage	Many-to-many link between User and Language
21	Income	Per-host income record

9. Maintenance Notes

- The script is **not idempotent**; drop the database before re-running.
- The ALTER TABLE block in Section D is essential; without it, the PlaceID foreign key in Accommodation is not created.
- The PaymentDate column is added after the Payment inserts through a separate ALTER TABLE, in order to preserve the linear execution order of the script.
- For very large datasets, indexes on Booking.GuestID, Booking.AccommodationID, Accommodation.HostID, Reservation.BookingID, and Reservation.PaymentID would accelerate the test-case joins.